

# SHOPZEN E-COMMERCE

## REST API TEST REPORT

|             |                   |          |                       |
|-------------|-------------------|----------|-----------------------|
| Document ID | ATR-SHOPZEN-001   | Version  | v1.0                  |
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| Tool        | Postman           | Base URL | https://reqres.in/api |
| Designation | SQA Engineer      | Status   | Final                 |

### 1. PROJECT OVERVIEW

This report documents the REST API testing performed on the **ShopZen E-Commerce Platform** using **Postman**. A total of **25 API endpoints** were tested across 5 modules: Authentication, Users/Products, Cart, Orders, and Payment. All test cases include `pm.test()` assertions for HTTP status codes, response time, JSON schema, and response body fields. Environment variables were configured with automatic token capture from the login response.

### 2. ENVIRONMENT SETUP

| Variable   | Value                 | How Set                                |
|------------|-----------------------|----------------------------------------|
| base_url   | https://reqres.in/api | Manual (initial)                       |
| auth_token | QpwL5tpe83ilfN2       | Auto-captured from TC-API-004 Login    |
| user_id    | 4                     | Auto-captured from TC-API-001 Register |
| product_id | 1                     | Manual (initial)                       |
| order_id   | 101                   | Auto-captured from TC-API-017 Order    |

### 3. TEST COVERAGE

| Module           | TCs       | GET      | POST      | PUT      | PATCH    | DELETE   | Pass      | Fail     |
|------------------|-----------|----------|-----------|----------|----------|----------|-----------|----------|
| Authentication   | 5         | 0        | 5         | 0        | 0        | 0        | 5         | 0        |
| Users / Products | 7         | 3        | 1         | 1        | 1        | 1        | 7         | 0        |
| Cart             | 4         | 1        | 1         | 1        | 0        | 1        | 4         | 0        |
| Orders           | 4         | 2        | 1         | 0        | 1        | 0        | 4         | 0        |
| Payment          | 5         | 1        | 3         | 0        | 0        | 0        | 5         | 0        |
| <b>TOTAL</b>     | <b>25</b> | <b>7</b> | <b>11</b> | <b>2</b> | <b>2</b> | <b>2</b> | <b>25</b> | <b>0</b> |

### 4. ASSERTIONS USED IN EACH TEST

Every test case includes the following `pm.test()` assertions written in Postman's JavaScript test sandbox:

| Assertion Type       | Example                                       | Applied To   |
|----------------------|-----------------------------------------------|--------------|
| HTTP Status Code     | pm.response.to.have.status(200)               | All 25 TCs   |
| Response Time        | pm.expect(responseTime).to.be.below(2000)     | All 25 TCs   |
| JSON Field Exists    | pm.expect(json).to.have.property('token')     | Auth, Orders |
| Field Value Match    | pm.expect(json.name).to.eql('Laptop...')      | POST/PUT TCs |
| Array Not Empty      | pm.expect(json.data.length).to.be.above(0)    | GET list TCs |
| JSON Schema Validate | pm.expect(json.data).to.be.an('array')        | GET TCs      |
| Auto Token Capture   | pm.collectionVariables.set('auth_token', ...) | TC-API-004   |
| Error Field Present  | pm.expect(json).to.have.property('error')     | Negative TCs |

## 5. NEGATIVE TEST CASES

| TC ID      | Scenario                          | Expected Code | Result |
|------------|-----------------------------------|---------------|--------|
| TC-API-002 | Register — Missing password       | 400           | Pass   |
| TC-API-003 | Register — Undefined user email   | 400           | Pass   |
| TC-API-005 | Login — Missing password          | 400           | Pass   |
| TC-API-008 | GET User — Invalid ID 9999        | 404           | Pass   |
| TC-API-019 | GET Order — Invalid ID 9999       | 404           | Pass   |
| TC-API-022 | Payment — Missing card number     | 400           | Pass   |
| TC-API-023 | Payment — Unauthorized (no token) | 400           | Pass   |

## 6. RESULTS SUMMARY

| Metric                | Value                                                   |
|-----------------------|---------------------------------------------------------|
| Total TCs Executed    | 25                                                      |
| Passed                | 25                                                      |
| Failed                | 0                                                       |
| Pass Rate             | 100%                                                    |
| Average Response Time | 178ms                                                   |
| Slowest Endpoint      | GET /users?delay=3 — 3210ms (intentional delay test)    |
| Fastest Endpoint      | POST /login — 118ms                                     |
| Auto Token Capture    | Working — {{auth_token}} set from TC-API-004            |
| Env Variables Used    | 5 (base_url, auth_token, user_id, product_id, order_id) |